

Eastern Bering Sea Pollock Model Development

Coordinating overview for the September 2026 Plan Team review

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August 26, 2026

Three working papers compare ways to build an eastern Bering Sea pollock model. This overview explains what each report contributes, where the results agree, how the approaches differ, and what guidance is requested from the Plan Team. The accepted ADMB assessment remains the basis for current management advice.

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i AI-assisted development

Artificial intelligence (AI) tools were used to aid document development and model implementation. AI-assisted material was reviewed and revised by the authors, who retain responsibility for the scientific content, code, analyses, interpretations, and conclusions.



COORDINATING OVERVIEW

1 Purpose of this overview

The September 2026 Plan Team package brings together three working papers on RTMB, SPoRC, and Rceattle. Each addresses a different development question. This page explains how the papers fit together, where their results agree, and what guidance is requested from the Plan Team. The papers are development studies rather than competing operational assessments.

! Important

The accepted ADMB assessment continues to provide current management advice. RTMB, SPoRC, and Rceattle are being used to check the model calculations, improve diagnostics, and evaluate options for future development. None is proposed as a replacement assessment for September 2026.

2 Review package

Report	Primary question	Role in the review
RTMB: main report	Can RTMB reproduce the base ADMB model, and what do its diagnostics and model alternatives show?	Main technical report and shared reference for comparing implementations
SPoRC: companion report	Can SPoRC reproduce the same base model and support different population-process structures?	Independent software comparison and continuing methods-development track
Rceattle: companion report	Can Rceattle reproduce the main assessment dynamics and provide a practical foundation for future operational development?	Leading candidate for future operational development, after the remaining diagnostic and review work

The [main RTMB report](#) provides the most detail on the shared data definitions, checks made using the same parameter values as ADMB, model fits, selectivity analyses, projections, retrospectives, and uncertainty diagnostics. The SPoRC and Rceattle reports show how the same assessment problem behaves in more general software frameworks.

2.1 Supporting analysis

Identifier	Analysis	Role in the review
S-1	Fishery selectivity alternatives (PDF)	Detailed RTMB comparison of the base, hierarchical double-logistic, and year-by-age AR1 fishery selectivity options; supports the selectivity discussion in the main RTMB report

This analysis supports the RTMB report; it is not a fourth modeling framework. The stable links allow all three reports to use the same equations, diagnostics, and summary data.

3 Where the results agree

The three reports reach several shared conclusions:

- All three comparisons use the base 2024 model with observed and predicted bottom-trawl survey age compositions evaluated over ages 1–15 and with the nominal multinomial sample sizes.
- When the same parameter values are used, the implementations closely reproduce the main ADMB population dynamics. This shows that the core model calculations have been transferred consistently.
- When each implementation estimates its own parameters, results differ because the parameterization, selectivity structure, and estimation methods differ.
- Time-varying fishery and survey selectivity remains an important source of sensitivity and some parameters remain weakly identified.
- Positive retrospective patterns require further evaluation before another framework is considered for operational use.
- Bayesian results remain exploratory until the chains mix adequately and effective sample sizes improve.

4 How the frameworks differ

Question	RTMB	SPoRC	Rceattle
What is it?	Direct RTMB representation of the base ADMB model	The same assessment expressed in a general single-region framework	The assessment expressed in a broader, extensible framework
Why is it useful?	Clear technical reference for checking calculations, diagnostics, and model changes	Independent check using different software and general population-process tools	Most promising route to a maintainable future operational model and broader extensions
What does it currently do well?	Detailed model-to-model checks, diagnostics, retrospectives, projections, and flexible selectivity development	Close reproduction of the base model, stable fits, and independent confirmation of the main dynamics	Close reproduction when parameter values are held fixed, an independently estimated comparison, general data and model architecture, and support for recruitment extensions

Question	RTMB	SPoRC	Rceattle
What work remains?	Retrospective bias and evaluation of candidate selectivity structures	Self-test failures, retrospective bias, and slow Bayesian mixing in the correlated survey-selectivity model	One weakly identified combination of annual selectivity parameters, retrospective bias, dependence on the observation model, and slow Bayesian mixing
Proposed use after this review	Keep as the technical reference and repeatable comparison test for later development	Continue as an independent methods and comparison track	Give priority as the operational-development candidate, after successful diagnostics, simulation testing, annual-update testing, and review

5 Proposed use of each framework

Each framework has a different development role. The proposed approach keeps those roles while concentrating operational-development effort on Rceattle.

1. **Retain ADMB for management advice.** Continue the accepted operational assessment while candidate frameworks are evaluated.
2. **Retain RTMB as the technical reference.** Use the base RTMB implementation to check likelihood calculations, state variables, projections, and proposed structural changes.
3. **Continue SPoRC as an independent comparison.** Use SPoRC to test whether conclusions depend on a particular software architecture and to evaluate general process-model developments.
4. **Prioritize Rceattle for operational development.** Resolve selectivity identification and retrospective behavior, complete observation-model and simulation tests, demonstrate a reproducible annual update, and return with a clearly defined candidate model.
5. **Make any transition through formal review.** Present a future operational candidate only after the fitted model, uncertainty analysis, projections, and update procedures pass agreed review criteria.

6 Questions for Plan Team discussion

The review would benefit from direction on the following questions:

1. Does the Plan Team agree that RTMB should serve as the implementation baseline while Rceattle receives priority for future operational development?
2. Which selectivity and observation-model sensitivities are essential before evaluating an operational Rceattle candidate?
3. What magnitude and pattern of retrospective bias would require model revision, additional uncertainty treatment, or rejection of a candidate?
4. Which simulation tests should be required to evaluate bias, estimability, and robustness under alternative recruitment and selectivity histories?
5. What evidence is needed to demonstrate that the annual data-update and projection procedures are ready for operational use?
6. At what review stage should posterior uncertainty be incorporated, given the current mixing limitations in candidate selectivity models?

7 Documents

HTML is the preferred format for reading and navigation. PDFs are provided for download, printing, and archiving.

- RTMB main report: [HTML \(preferred\)](#); [PDF](#)
- Fishery selectivity analysis: [HTML \(preferred\)](#); [PDF](#)
- SPoRC companion report: [HTML \(preferred\)](#); [PDF](#)
- Rceattle companion report: [HTML \(preferred\)](#); [PDF](#)